REVISION NOTES

Plus One Higher Secondary (Class 11) Physics - High-Yield Revision Notes & Formula Book

Kinematics, Laws of Motion, Work-Energy Theorem, Gravitation & Thermodynamics

Prescribed Institution / Board:	Directorate of Higher Secondary Education (DHSE Kerala)
Degree / Academic Level:	Plus One (Class 11 Science Stream)
Subject / Module Code:	Physics (Subject Code: 501)
Curriculum Academic Year:	2026-2027
Author / Academic Department:	Higher Secondary Physics Resource Group
Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Chapter 1 & 2: Motion in a Plane & Laws of Motion

1.1 Projectile Motion Formulations

A projectile launched with initial velocity u at angle Î¸ to the horizontal experiences constant downward acceleration g and zero horizontal acceleration.

Maximum Height: H = (u^2 * sin^2 Î¸) / (2g)

Time of Flight: T = (2u * sin Î¸) / g

Horizontal Range: R = (u^2 * sin 2Î¸) / g. Range is maximum at launch angle Î¸ = 45 degrees.

- â€¢ Newtons Second Law: F = dp/dt = m * a (for invariant mass).
- â€¢ Law of Conservation of Linear Momentum: Initial total momentum equals final total momentum when external net force equals zero.
- â€¢ Banking of Roads without friction: tan Î¸ = v^2 / (r * g).

1.2 Work, Energy and Power Theorem

Work-Energy Theorem states that the net work done by all forces on a body equals the change in its kinetic energy: W_net = Î”K = K_final - K_initial.

Conservative Force criterion: Work done in a closed loop vanishes: âˆ® F â€¢ dr = 0.

Chapter 3: Thermodynamics & Kinetic Theory

3.1 Laws of Thermodynamics

Zeroth Law: Establishes temperature as a fundamental thermal equilibrium property.

First Law: Conservation of Energy: dQ = dU + dW, where dW = P * dV.

Isothermal Process: Temperature remains constant: PV = constant, W = nRT ln(V2/V1).

Adiabatic Process: Zero heat exchange (dQ = 0): PV^Î³ = constant, W = [nR(T1 - T2)] / (Î³ - 1).

Carnots Heat Engine: Maximum theoretical efficiency Î· = 1 - (T_sink / T_source).